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Diversity-oriented synthesis of functional poly(trimethylene *p*-phenylene) through acetoacetate-di(bromomethyl)benzene polymerization

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ABSTRACT

Diversity-oriented polymerization is a powerful synthetic approach for generating diversified polymers with novel structures and advanced functionalities. However, realizing diversity-oriented synthesis in two-component (bimolecular) polymerizations presents a significant challenge. Herein, the two-component polymerizations of alcohol-derived acetoacetates and di(bromomethyl)benzenes were reported to facilely fabricate 14 poly(trimethylene *p*-phenylene)s showcasing diverse, well-defined structures, high molecular weights (with M_w's reaching 27.9 kDa), and impressive yields (up to 92 %). These polymers exhibit tailored thermal ($T_{d,5\%}$ exceeding 200 °C) stability and glass transition temperatures ranging from −18 to 50 °C, with the tetraphenylethene-containing variant displaying distinctive aggregation-induced emission (AIE) properties. Notably, the poly(trimethylene *p*-phenylene) bearing *tert*-butyl ester groups could be facilely post-modified through sequential hydrolysis and decarboxylation of acetoacetate groups. Thus, this work not only establishes a new diversity-oriented two-component polymerization but also yields a range of poly(trimethylene *p*-phenylene)s with customizable structures and properties.

1. Introduction

Diversity-oriented synthesis [1] of functional polymers plays a crucial role in the development of innovative materials with desirable properties [2]. The number, type, and placement of functional groups are key characteristics that define the properties and utility of macromolecular materials [3,4]. Polymerizing functionalized monomers is the most direct route toward a polymer with desired side-chain functionality [5] Over the past decade, diverse multi-component polymerization [6–8] (MCP) and multi-component tandem polymerizations [9,10] have been successfully demonstrated, showcasing the design of a spectrum of functional polymers. However, achieving diversity-oriented synthesis in two-component (bimolecular) polymerizations presents a significant challenge. In this context, bifunctional monomers must incorporate another functional group compatible with the polymerization reaction, leading to intricacies in monomer design and synthesis [11]. As a result, only a limited number of examples (Scheme 1), such as amino-epoxy [12,13], amino-acrylate [14–16] /divinyl sulfone [17,18], and thiol-yne [19,20] polymerizations, have been reported. Given the advantages of a simple process and the accessibility of raw materials, there is a strong impetus to explore and develop diversity-oriented two-component polymerizations.

Active methylene compounds (AMCs) [21,22], characterized by their structural diversity and unique reactivity, serve as valuable building blocks in various reactions such as alkylation [23], aldol condensation [24], Knoevenagel condensation [25], Michael addition [26], etc. With diverse electron-withdrawing groups, AMCs include compounds like acetoacetates, malonates, 1,3-diketones, malononitrile, cyanoacetates, cyanoacetamides, barbituric acid, Meldrum's acid, 2-arylacetonitriles and thiazolidinedione [27]. However, the great advantage of AMCs has not been fully utilized in polymer science until now. For example, dynamic condensation products of acetoacetate-amine [28,29] and Knoevenagel adducts of cyanoacetates [30] /acetoacetates [31] have been used to prepare malleable thermosets. Additionally, the reversible coupling of amines and thiols via a Meldrum's acid [32]/1,3-Indandione [33]-based conjugate acceptor has been harnessed to control the morphological transformations and degradation of polymers. Furthermore, the Michael addition reactions involving acetoacetates and acrylates are valuable for preparing cross-linked materials [34,35].

In recent years, significant efforts have been directed towards enhancing the structural diversity of AMC-derived polymers. Leveraging the versatile reactivity of AMCs, several polymerization strategies have

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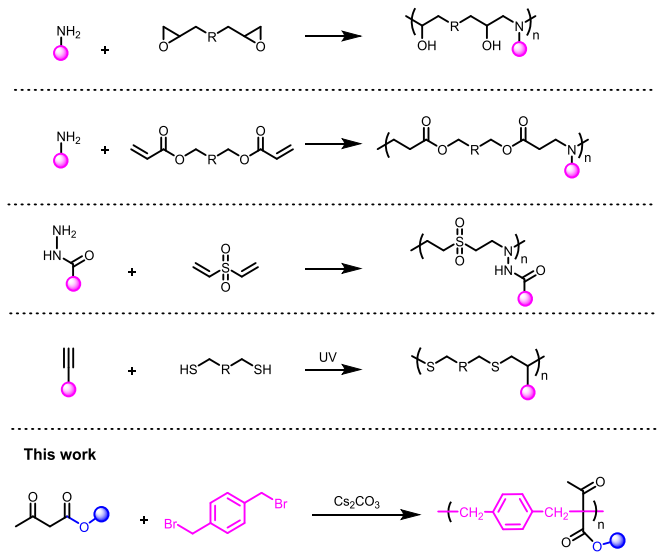

Scheme 1. Comparison of previously reported diversity-oriented two-component polymerizations with this work.

been developed, such as the AMC-carbon disulfide-dihalohydrocarbon MCP [36], Meldrum's acid-indole-aldehyde MCP [37], AMC-acrylate polyaddition [38], AMC-dihalogen polyalkylation [39], and 1,3-dicarbonyl compound-di(bromomethyl)benzene [40] polycondensation. Furthermore, AMCs have been employed for post-modification to synthesize functional polymers. For instance, Hantzsch [41] and Beginelli [42] multicomponent reactions based on acetoacetates have been integrated with radical polymerization for high-throughput combinatorial synthesis. Additionally, reversible acetoacetate [43]/β-triketone [44]-based transamination and Knoevenagel reactions of Meldrum's Acid [45] have been utilized for dynamic post-polymerization modifications. These achievements underscore the significant potential of AMC building blocks in the creation of structurally diverse polymers.

Herein, we present the pioneering utilization of acetoacetate- di(bromomethyl)benzene polycondensation as a versatile and robust diversity-oriented two-component polymerization. The acetoacetate monomer exhibits dual functionality: its active methylene group engages in polycondensation with di(bromomethyl)benzene to form the polymeric chain, while its ester group facilitates the incorporation of side groups. Various acetoacetate monomers can be readily synthesized by reacting diketene with corresponding alcohols, which are abundantly available and structurally diverse. With exceptional tolerance towards functional groups, this polycondensation can be smoothly conducted at 25 °C in air, yielding 14 poly(trimethylene *p*-phenylene) polymers with diverse structures, high molecular weights, and excellent yields. These polymers demonstrate adjusted thermal stability and glass transition temperatures, with the tetraphenylethene-containing variant exhibiting unique aggregation-induced emission (AIE) characteristics. Thanks to the distinctive reactivity of the acetoacetate group, the poly(trimethylene *p*-phenylene) bearing *tert*-butyl ester groups can be transformed into a novel carbonyl-functionalized polymer through sequential hydrolysis and decarboxylation reactions in a one-pot process.

2. Results and discussions

2.1. Polymerizations of acetoacetates and di(bromomethyl)benzenes

To develop the polymerizations of acetoacetates and di(bromomethyl)benzenes, commercially available ethyl acetoacetate **1a** and 1,4-di(bromomethyl)benzene **2a** were selected as the representative monomers (Fig. 1). With the equimolar equivalents of **1a** and **2a** and a high monomer concentration of [**1a**] = 1.0 M, the polymerization was carried out in the presence of Cs_2CO_3 in *N*,*N*-dimethylformamide (DMF) at 25 °C for 24 h, giving a soluble poly(trimethylene *p*-phenylene) product with a weight-average molecular weight (M_w) of 11.8 kDa in 81 % yield (**entry 1 in** Table 1).

The solvent, time, catalyst, and monomer concentration of the polymerization were then optimized subsequently, considering the M_w values and yields of the polymer product. Elongation of the reaction time dramatically increased the M_w and yield, producing a polymer with the M_w value of 18.6 kDa in 92 % yield after 72 h (**entry 2 in** Table 1 **and Fig. S1**). The monomer concentration was crucial to this polymerization. At a concentration of 0.5 M for both [**1a**] and [**2a**], a polymer product with the M_w value of 17.9 kDa was obtained in 85 % yield (**entry 3 in** Table 1). However, increasing the monomer concentration to 1.5 M and 2.0 M resulted in decreased M_w values of 14.1 kDa and 14.9 kDa,

Table 1
Polymerizations of **1a** and **2a**.[a]

Entry	Solven	[1a]/(M)	Yield (%)[b]	M_w (kDa)[c]	Đ[c]
1[d]	DMF	1.0	81	11.8	1.62
2	DMF	1.0	92	18.6	1.67
3	DMF	0.5	85	17.9	2.21
4	DMF	1.5	82	14.1	1.67
5	DMF	2.0	64	14.9	2.13
6	DIXO	1.0	90	9.9	1.70
7	DMSO	1.0	80	25.5	4.10
8[e]	DMF	1.0	<50	0.3	1.00
9[f]	DMF	1.0	<50	0.7	1.50

[a] Carried out at 25 °C in air for 72 h ([1a]/[2a] = 1.0, [Cs_2CO_3]/[1a] = 2.2).
[b] Determined yields of the product after methanol precipitation.
[c] Determined by GPC in tetrahydrofuran (THF), calibrated with polystyrene standards.
[d] For 24 h.
[e] The catalyst is Et_3N. [f] The catalyst is K_2CO_3.
[f] The catalyst is K_2CO_3.

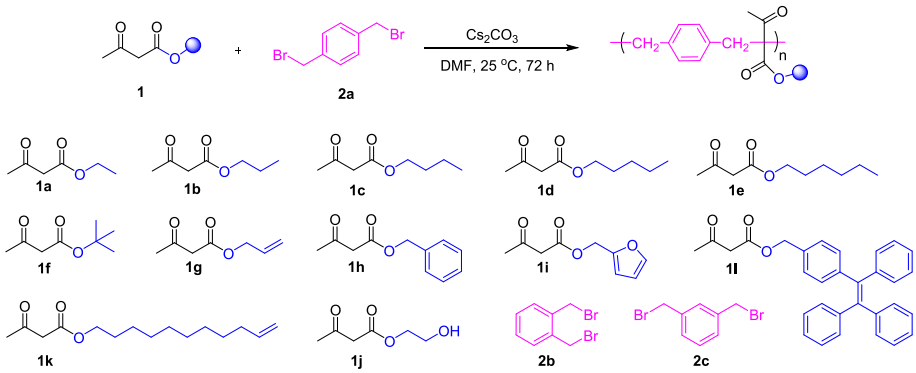

Fig. 1. Polymerizations of acetoacetates and di(bromomethyl)benzenes. Investigated scope of monomers.

respectively (**entries 4 and 5 in** Table 1). In the presence of the byproduct cesium bromide, high monomer concentration increased the viscosity of the polymerization mixture, thus limiting the reaction extent of polymer end groups. Notably, DMF and dimethyl sulfoxide (DMSO) were proved to be the superior solvents for the polymerization than 1,4-dioxane, probably due to the favorable solubility of the cesium salt (**entries 6 and 7 in** Table 1). In contrast to Cs_2CO_3, triethylamine (Et_3N) and K_2CO_3 exhibited lower efficiency in catalyzing this polymerization reaction, indicating the critical role of appropriate alkalinity in the system's reactivity (**entries 8 and 9 in** Table 1). Consequently, we chose a monomer concentration of 1 M, DMF as the solvent, Cs_2CO_3 as the catalyst, a reaction time of 72 h, and 25 °C as the optimal polymerization conditions for further investigation (Scheme S1).

The well-defined structure of the poly(trimethylene *p*-phenylene) was confirmed by their ^{1}H and ^{13}C NMR spectra. To facilitate the structural characterization of polymer products, model compound **3** was synthesized by the condensation between ethyl acetoacetate (1 eq.) and benzyl bromide (2 eq.) under the identical reaction conditions (Fig. 2 **and** Scheme S2). In the ^{1}H NMR spectra shown in Fig. 3, the methyl protons (H^1) adjacent to the carbonyl group of **1a** resonated at 2.27 ppm shifted to 1.98 ppm (H^4) and 1.90 ppm (H^6) in the spectra of **3** and **P1**, respectively. The active methylene protons (H^2) between the two carbonyl groups of **1a** resonated at 3.44 ppm disappeared in the spectra of **3** and **P1**, proving the complete conversion of **1a**. Moreover, the proton resonances of the benzyl methylene group at 4.52 ppm shifted to 3.24 ppm (H^5) and 3.12 ppm (H^7) in the spectra of **3** and **P1**, respectively. Meanwhile, in the ^{13}C NMR spectra of **3** and **P1**, the characteristic acetoacetate carbon peaks of monomer **1a** at 29.42 ppm and 49.80 ppm, shifted to 28.06 ppm and 66.04 ppm, respectively, while the benzyl carbon peak of monomer **2a** at 32.96 ppm shifted to 39.73 ppm. All these results confirmed that well-defined poly(trimethylene *p*-phenylene) was generated by the polymerizations of ethyl acetoacetate and di (bromomethyl)benzenes (**Figs. S2-S5**).

2.2. Generality of the polymerization

In order to test the robustness and universality of this polymerization, a diverse array of monomers was examined under the optimized polymerization conditions (Fig. 4). Acetoacetate monomers can be facilely synthesized by the reaction of the corresponding alcohols and diketene at 25 °C (Scheme S3). A series of acetoacetate monomers with inert groups (ethyl, propyl, *n*-butyl, *n*-pentyl, *n*-hexyl, *tert*-butyl, benzyl), reactive groups (allyl), chromophoric group (4-(1,2,2-triphenylethenyl)-benzyl), and biomass-derived groups (furfuryl, hydroxyethyl, 10-undecen-1-yl) were investigated in this polymerization (**Figs. S6-S10**).

The polymerizations of all 14 combinations of acetoacetates and di (bromomethyl)benzenes were observed to occur spontaneously, yielding polymers with high M_w values (up to 27.9 kDa) in impressive yields (up to 92 %). Substituting 1,4-di(bromomethyl)benzene with 1,2- or 1,3-di(bromomethyl)benzene resulted in the production of **P12** and **P13** with lower M_w values and reduced yields, indicating 1,4-di(bromomethyl)benzene with less steric hindrance was critical for this polymerization.

The ^{1}H NMR spectra of all 14 poly(trimethylene *p*-phenylene)s

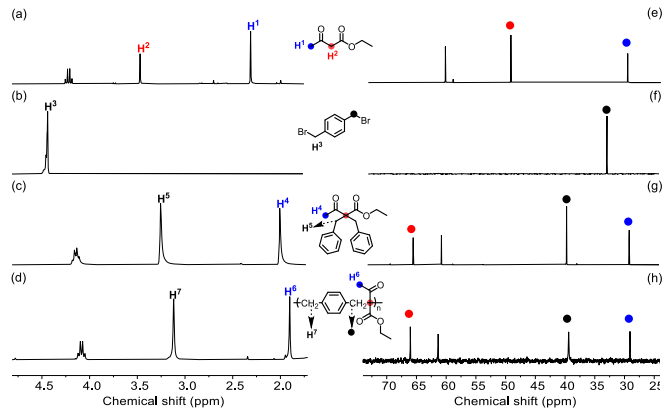

Fig. 3. ^{1}H NMR spectra of (a) **1a**, (b) **2a**, (c) **3**, and (d) **P1** and ^{13}C NMR spectra of (e) **1a**, (f) **2a**, (g) **3**, and (h) **P1** in $CDCl_3$.

showcased the emergence of the characteristic methyl signals at 1.90–2.00 ppm, while the ^{13}C NMR spectra exhibited the characteristic benzyl methylene signals at 39.35–39.42 ppm, confirming their chemical structures (**Figs. S11 − S36**).

The structural diversity inherent in these poly(trimethylene *p*-phenylene)s confers them with customizable properties, including thermal stability and thermally induced phase transitions. Most polymers exhibited high $T_{d,5\%}$ (decomposition temperature at 5 % weight loss) of exceeding 200 °C, as determined by thermogravimetric analysis (TGA, Figs. 4**b and** S37–S50). **P6** demonstrated a low $T_{d,5\%}$ of 209 °C due to the incorporation of easy-to-break *tert*-butyl groups, while **P12** and **P13** exhibited the lowest $T_{d,5\%}$ of around 170 °C possibly due to their reduced molecular weights.

According to differential scanning calorimetry (DSC) measurements, all the poly(trimethylene *p*-phenylene)s exhibited apparent glass transitions with the glass transition temperature (T_g) values ranging from −18 to 50 °C (Figs. 4**c and** S51–S64). The presence of bulky substitutes without hydrogen bonding prevented these polymers from crystalline. Generally speaking, polymers with aromatic rings or hydroxyl side groups exhibit higher glass transition temperatures than those with fatty side groups, due to the presence of stronger intermolecular forces. Interestingly, **P6** featuring *tert*-butyl groups exhibited the highest T_g values of 50 °C, attributed to the restriction of segmental movement by steric hindrance. Therefore, the polymerizations of acetoacetates and di (bromomethyl)benzenes provides a robust and facile platform to synthesize functional polymers with diverse structures.

2.3. Aggregation-induced emission

Benefiting from good functional group tolerance, this polymerization reaction facilitates the tailored design of polymers with specific functions. Here, we can endow the polymer with unique fluorescence properties by introducing tetraphenylethene functional groups. In contrast to aggregation-caused quenching, aggregation-induced emission (AIE) describes the phenomenon where a dye shows enhanced emission upon transitioning from isolated molecular states to

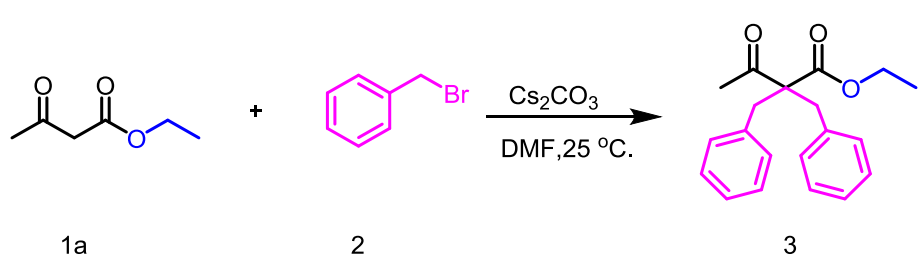

Fig. 2. Synthesis of model compound **3**.

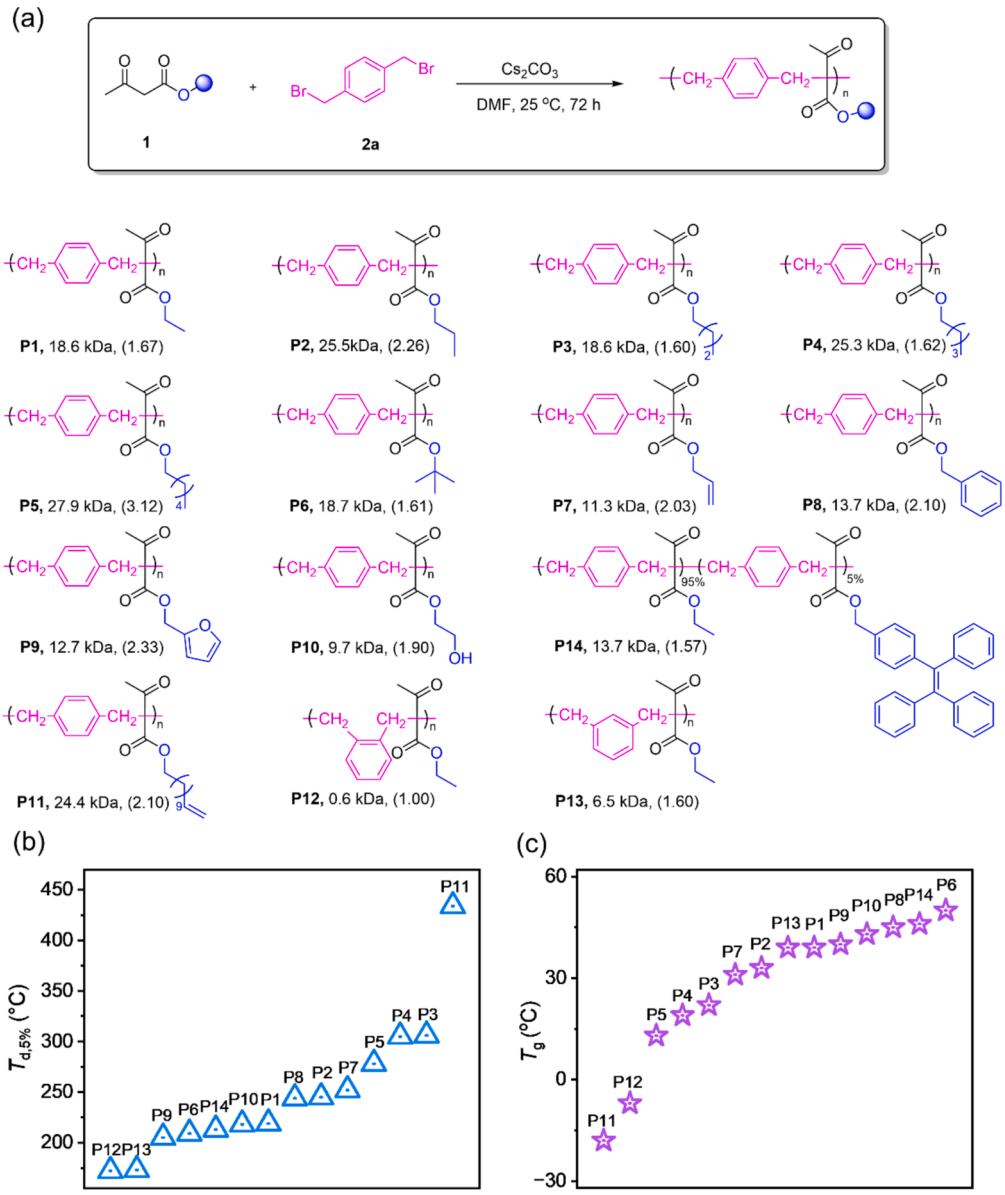

Fig. 4. (a) Chemical structures and polymerization results of polymers **P1**-**P14**. M_w and $Đ$ values are determined by GPC in THF calibrated with polystyrene standards. $T_{d,5\%}$ (b) and T_g (c) values of **P1** to **P14**.

aggregated states[46]. Tetraphenylethene is a well-known AIE molecule. To this end, we designed and synthesized poly(trimethylene *p*-phenylene) **P14** containing tetraphenylethene groups through the polymerization of ethyl acetoacetate (0.95 eq.), [4-(1,2,2-triphenylethenyl)phenyl]methyl acetoacetate (0.05 eq.), and *p*-bromomethyl benzene (1 eq.). Its chemical structure and molecular weight were characterized by ^{1}H NMR and GPC (as shown in **Fig. S35**).

Subsequently, the fluorescence characteristics of polymer **P14** in various solvents were examined, and the results are shown in Fig. 5. Notably, in THF solvent, **P14** demonstrated complete solubility; under 366 nm light excitation, the solution emitted weak fluorescence with a maximum emission wavelength of 410 nm. Given the hydrophobic nature of the polymer, we further examined its fluorescence behavior in THF/water mixtures at varying ratios to investigate the impact of polymer aggregation on its luminescent properties. Fig. 5 shows that the fluorescence intensity continued to increase with increasing water content, reaching its peak at 90 % water content. Compared to the value observed in THF solution, this maximum fluorescence intensity represents a significant enhancement of 72-fold. These results indicate that polymer **P14** exhibited unique AIE[47]. In THF/water mixed solvents, the decrease in polymer solubility and the resulting aggregation hinders the free rotation of tetraphenylethene, thereby enhancing its photoluminescence by restricting non-radiative decay pathways. It is worth noting that the increase of water content also led to a gradual red-shift of the maximum emission wavelength from 410 nm to 470 nm, which may be due to the transition of polymeric chromophores from dispersed molecular states to aggregation-constrained states.

2.4. Post-polymerization modification via tandem hydrolysis and decarboxylation reactions

Post-polymerization modification is an effective strategy for achieving polymer conversion and expanding structural diversity[48]. The poly(trimethylene *p*-phenylene) containing reactive functional groups designed here can undergo subsequent transformative processes. For instance, **P9** with furan functionalities can engage in efficient Diels-Alder reactions; while **P10** containing hydroxyl groups is reactive with isocyanates; **P7** and **P11**, featuring double bonds, are conducive to thiol-

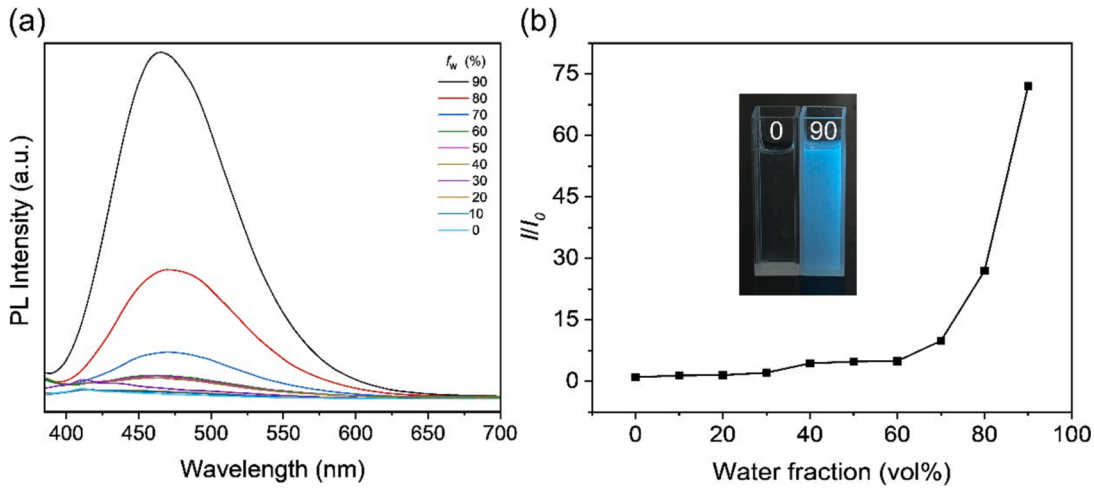

Fig. 5. (a) **P14** in THF/water mixtures with different water contents. Concentrations: 1.3 × 10^{-3} M. Excitation wavelength: 366 nm. (b) Plot of relative fluorescence intensity versus water fraction in THF/water mixtures, where I = peak intensity in water mixtures and I_0 = peak intensity in pure THF. The inset in panel b: photographs of **P14** in pure THF and a THF/water mixture with 90 % water and taken under illumination of a hand-held UV lamp.

ene click reactions.

Polymer **P6** containing *tert*-butyl groups was taken as an example to demonstrate the potential for post-polymerization modification of these polymers. This polymer is characteristic of substituted *tert*-butyl acetoacetate structures. Under acidic conditions, *tert*-butyl esters are easily hydrolyzed to carboxyl groups, achieving the transformation of this characteristic functional group into a substituted acetoacetic acid structure. The latter is prone to decarboxylation under acidic conditions, resulting in the formation of substituted ketone structures. Therefore, by utilizing the unique hydrolysis and decarboxylation reactions of acetoacetates, we envision that acid treatment can achieve the synthesis of carbonyl functionalized polymers starting from polymer **P6** (Fig. 6**a**).

To test this hypothesis, polymer **P6** was dissolved in a dichloromethane solution containing trifluoroacetic acid (3 % v/v), and its hydrolysis process was monitored by ^{1}H NMR measurements. As shown in Fig. 6**b**, the proton peak of the *tert*-butyl group at 1.25 ppm gradually disappeared over time, indicating that the occurrence of the hydrolysis reaction. Particularly noteworthy was the gradual disappearance of the characteristic acetoacetate group's proton peak at 1.97 ppm, concomitant with the incremental emergence of a peak at 1.69 ppm. The latter

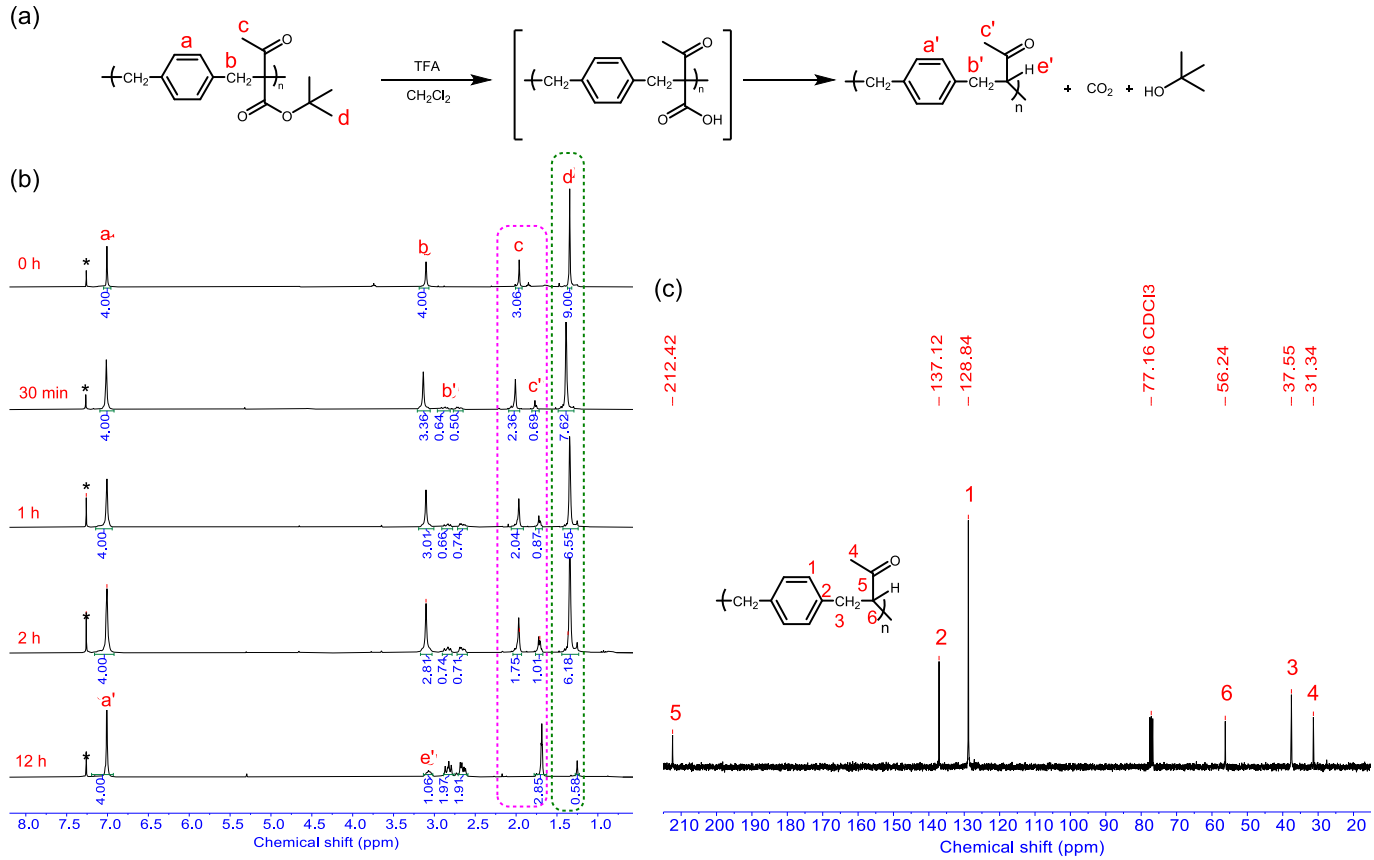

Fig. 6. (a) The tandem hydrolysis and decarboxylation of **P6**. (b) ^{1}H NMR of hydrolysis process at different time. (c) ^{13}C NMR of hydrolysis products.

corresponded to the acetyl group of the polymer product after hydrolysis and decarboxylation (Fig. S65). The ^{13}C NMR in Fig. 6**c** provided further evidence: the initial polymer **P6** possessed two carbonyl carbon atoms at 170.56 ppm and 205.58 ppm, attributable to acetoacetate groups; following the hydrolytic event, only one carbonyl carbon atom remained at 212.42 ppm in the polymer product, indicating the occurrence of decarboxylation. Concurrently, the remained carbon peak positions align correspondingly with the intended product. GPC analysis showed a significant decrease in M_w from 18.8 kDa to 7.6 kDa, which is consistent with the expected results after decarboxylation (Fig. S66). All these data demonstrates that polymer **P6** could undergo sequentially hydrolysis and decarboxylation reactions, transforming into the envisioned carbonyl functionalized polymer.

3. Conclusion

In this work, we successfully established a diversity-oriented two-component polymerization between alcohol-derived acetoacetates and di(bromomethyl)benzenes. This polymerization enjoys the advantages of readily available monomers, simple operation, high efficiency, mild conditions, and excellent functional group tolerance. By adjusting the chemical structures of acetoacetate monomers, 14 poly(trimethylene *p*-phenylene)s with inert, reactive, chromophoric, and biomass-derived groups were obtained with high M_w values (up to 27.9 kDa) in impressive yields (up to 92 %) under ambient conditions. Tailored thermal stability ($T_{d,5\%}$, exceeding 200 °C) and glass transition temperatures ranging from −18 to 50 °C were observed, with a tetraphenylethene-containing variant showcasing distinctive AIE properties. Notably, facile post-modification of the poly(trimethylene *p*-phenylene) carrying *tert*-butyl ester groups was demonstrated through sequential hydrolysis and decarboxylation of acetoacetate functionalities. This work not only establishes a novel diversity-oriented two-component polymerization pathway but also produces a variety of poly(trimethylene *p*-phenylene) polymers with customizable properties and structures.

CRediT authorship contribution statement

Mengdie Guo: Investigation, Formal analysis, Writing – original draft. **Shengyu Feng:** Supervision. **Linlin Wang:** Conceptualization, Writing – review & editing. **Lei Li:** Writing – review & editing, Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary data

Supporting Information is available and includes general experimental considerations, detailed procedures, and characterization spectra. Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113654.

Data availability

Data will be made available on request.

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